WORKED SOLUTION

STEP I 2001, Question 11

Mechanics • Elastic Strings and Energy

STEP I 2001, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

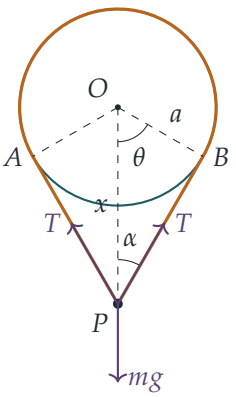
step.maths.org/questions/01-s1-q11

The key idea

Once the particle pulls the band down, the band stays on the cylinder over most of its length and leaves it *tangentially* on each side, running straight to the particle. So everything comes down to the geometry of two tangents from an external point to a circle. Because the cylinder is smooth, the tension is the same all the way round the band, so Hooke's law with the *total* extension gives it. Then part (i) is an energy question (the particle is momentarily at rest at its lowest point) and part (ii) is a force question (maximum speed means zero acceleration).

Solution

We model the particle as a point mass, the band as light and elastic (obeying Hooke's law), and the cylinder as smooth. Let O be the centre of the cross-section and suppose the particle P is a distance x below O, with $x \geq a$. The band leaves the circle at points A and B, where PA and PB are tangents.



The geometry and the tension

Triangle OAP has a right angle at A (a tangent is perpendicular to the radius), so

$$PA = PB = \sqrt{x^2 - a^2}, \qquad \cos\theta = \frac{a}{x}, \qquad \sin\alpha = \frac{a}{x},$$

where $\theta = \angle AOP$ and $\alpha = \angle APO$, the angle each straight piece of band makes with the vertical. The band is in contact with the circle everywhere except on the arc AB of angle 2θ nearest P, so its length is

$$L = (2\pi a - 2a\theta) + 2\sqrt{x^2 - a^2}.$$

The natural length is $2\pi a$, so the extension is

$$e(x) = 2\left(\sqrt{x^2 - a^2} - a\theta\right), \qquad \theta = \arccos\frac{a}{x}.$$

Since the cylinder is smooth, there is no friction along the band, so the tension T is the same throughout, and by Hooke's law $T = \dfrac{\lambda e}{2\pi a}$.

At $x = 2a$ we have $\cos\theta = \frac{1}{2}$, so $\theta = \frac{\pi}{3}$, $\alpha = \frac{\pi}{6}$ and $\sqrt{x^2 - a^2} = \sqrt{3}\,a$. Hence

$$e(2a) = 2\left(\sqrt{3}\,a - \frac{\pi a}{3}\right) = \frac{2a}{3}\left(3\sqrt{3} - \pi\right).$$

Part (i)

At release the particle is at the bottom of the cylinder, $x = a$, the band is unstretched and the particle is at rest. At $x = 2a$ it is at rest again (it falls no further), so it has no kinetic energy at either moment. The cylinder is smooth, so no energy is lost, and the loss of gravitational potential energy equals the gain in elastic energy. The particle has fallen a distance $2a - a = a$, and the elastic energy of a band of natural length $2\pi a$ with extension e is $\dfrac{\lambda e^2}{2 \cdot 2\pi a}$. So

$$mga = \frac{\lambda}{4\pi a} \cdot \frac{4a^2}{9}\left(3\sqrt{3} - \pi\right)^2 = \frac{\lambda a\left(3\sqrt{3} - \pi\right)^2}{9\pi},$$

which gives

$$\lambda = \frac{9\pi mg}{\left(3\sqrt{3} - \pi\right)^2},$$

as required.

Ask yourself Is it fair to use the elastic energy formula for the whole band, when part of it is wrapped round the cylinder?

Yes. Because the cylinder is smooth, the tension, and so the stretch per unit length, is the same along the whole band, wrapped or not. The band behaves exactly like a single elastic string of natural length $2\pi a$ with total extension e, and that is what the formula $\lambda e^2/(2l)$ describes.

Part (ii)

The speed is greatest when the acceleration is zero, that is, when the resultant force on the particle is zero. Horizontally the two tensions balance by symmetry. Resolving vertically at P,

$$2T\cos\alpha = mg.$$

At $x = 2a$, $\cos\alpha = \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$, so $\sqrt{3}\,T = mg$, giving $T = \dfrac{mg}{\sqrt{3}}$. By Hooke's law with the extension found above,

$$T = \frac{\lambda}{2\pi a}\cdot\frac{2a}{3}\left(3\sqrt{3}-\pi\right) = \frac{\lambda\left(3\sqrt{3}-\pi\right)}{3\pi}.$$

Equating the two expressions for T,

$$\lambda = \frac{3\pi mg}{\sqrt{3}\left(3\sqrt{3}-\pi\right)} = \frac{\sqrt{3}\,\pi mg}{3\sqrt{3}-\pi}.$$

Ask yourself How do we know this is a maximum of the speed and not just a stationary point?

As x increases, e increases (each piece of band gets longer faster than the arc $a\theta$ shrinks) and $\cos\alpha = \sqrt{1-a^2/x^2}$ increases, so the upward force $2T\cos\alpha$ increases steadily from 0. The net downward force is positive above $x = 2a$ and negative below it, so the particle speeds up until $x = 2a$ and slows down afterwards.

Checks

- A neat consistency check: differentiating L gives $\dfrac{\mathrm{d}L}{\mathrm{d}x} = \dfrac{2x}{\sqrt{x^2-a^2}} - \dfrac{2a^2}{x\sqrt{x^2-a^2}} = \dfrac{2\sqrt{x^2-a^2}}{x} = 2\cos\alpha$. So the rate of change of elastic energy, $T\dfrac{\mathrm{d}L}{\mathrm{d}x}$, equals $2T\cos\alpha$, the upward force. The energy method of (i) and the force method of (ii) agree.
- Numerically, $\lambda \approx 6.70\,mg$ in (i) and $\lambda \approx 2.65\,mg$ in (ii). The band in (ii) is weaker, which makes sense: there the particle is still moving at $x = 2a$ and carries on past it, whereas in (i) a stiffer band has already brought it to rest.
- Stepping the equation of motion $m\ddot{x} = mg - 2T\cos\alpha$ numerically from rest at $x = a$ gives a lowest point at $x = 2.0000a$ for the first value of λ, and maximum speed at $x = 2.0000a$ for the second.

Common slips

- Forgetting that some of the band leaves the cylinder: the extension is $2\sqrt{x^2-a^2}$ *minus* the lost arc $2a\theta$, not just the length of the two straight pieces.

- Measuring the fall from the axis instead of from the release point: the particle falls a distance a, not $2a$.
- Using $\sin\alpha$ instead of $\cos\alpha$ when resolving vertically; α is measured from the vertical, and $\sin\alpha = a/x$.

Beyond STEP

The check that $\mathrm{d}(\text{elastic energy})/\mathrm{d}x$ equals the restoring force is an instance of the principle of virtual work: in a conservative system the force in any direction is minus the rate of change of potential energy in that direction. This viewpoint, where you describe a system by one coordinate and its potential energy, is the starting point of Lagrangian mechanics, which you will meet in the first or second year of a university mechanics course. For a careful treatment of energy, forces and potentials, try Kleppner and Kolenkow, *An Introduction to Mechanics*.

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